

Name: Faizan Khurram

10 Pearls Project Report
(AQI Predictor)

Email: faizan.khurram@hotmail.com

Github Repo Link:
<https://github.com/faizankhurram/AQI-Predictor>

1. Problem Statement

Karachi faces recurring poor air quality from traffic, industry, and seasonal dust. Residents need more than a single current reading i.e they need a short-horizon outlook to plan outdoor activity and early response. This project delivers an automated 3-day US AQI forecast (+24h, +48h, +72h) for Karachi using hourly Open-Meteo data, a cloud feature store, daily model retraining, and a public Streamlit dashboard.

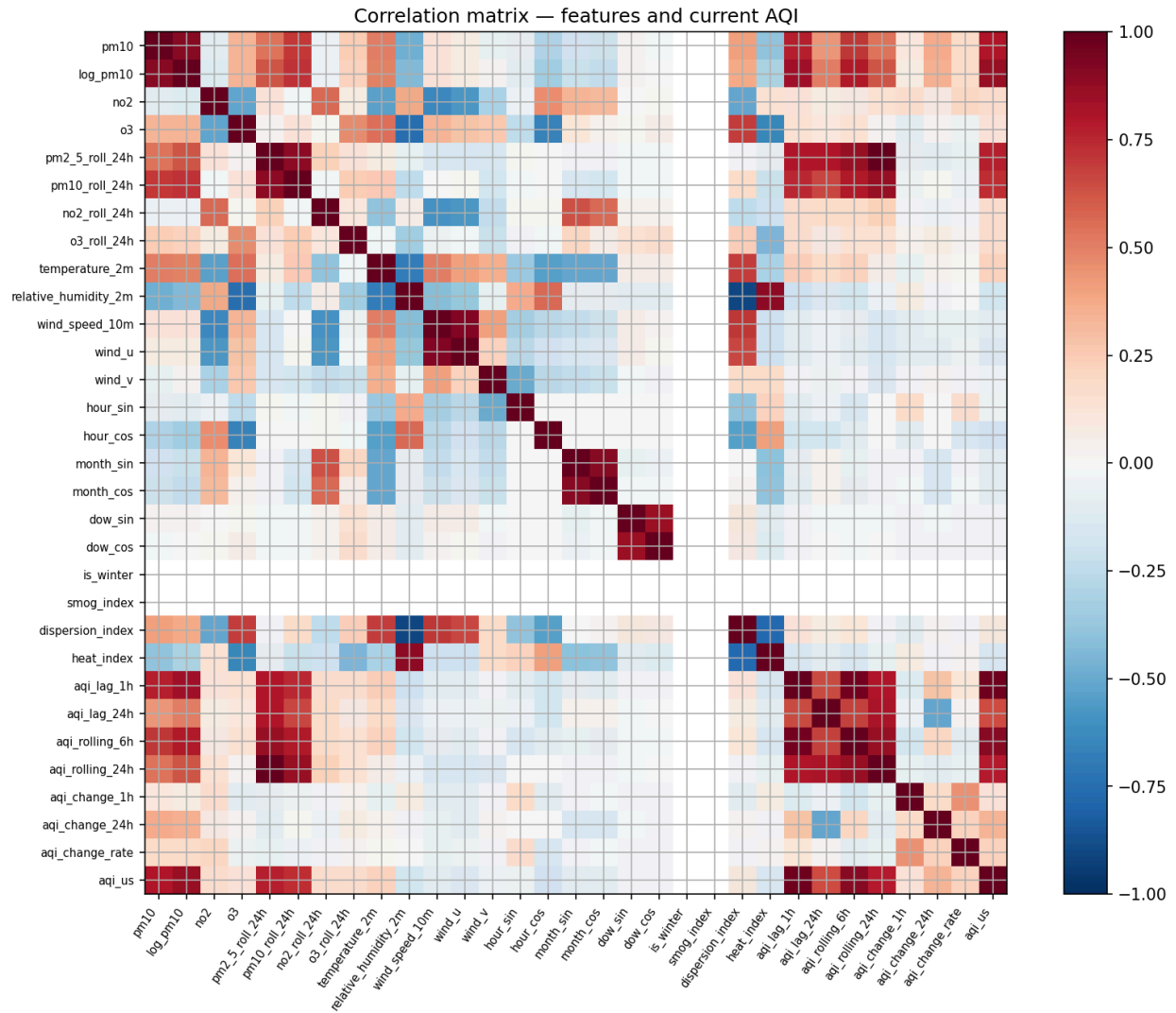
2. Preprocessing and Feature Engineering:

Raw PM2.5 is calibrated ($\times 1.42$) for South Asian sensor bias; US AQI is recomputed with 2024 EPA breakpoints. Feature engineering includes 24h rolling means per pollutant, cyclic hour/day/month encodings, wind U/V components, AQI lags and change rates, and interaction indices (smog, dispersion, heat). The main targets are future AQI at +24h, +48h, and +72h via backward shift. Incomplete rows from lags/rolls are dropped and volatile columns are winsorized on the train split only.

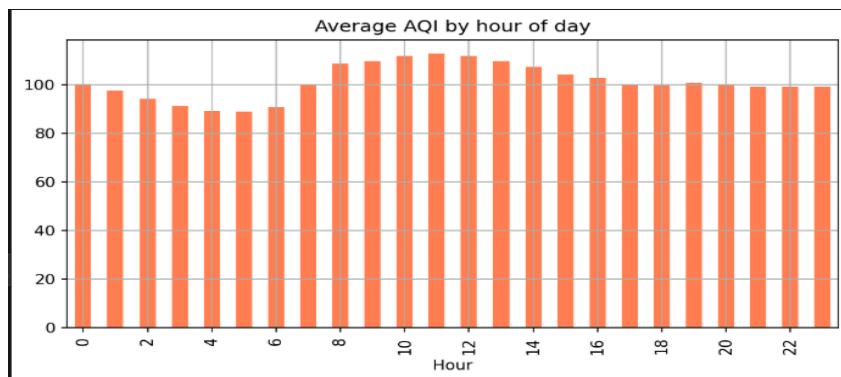
3. Exploratory Data Analysis Findings:

The following are some of the important highlights and visuals from the performed EDA:

(i): PM2.5 dominates US AQI (correlation > 0.95).

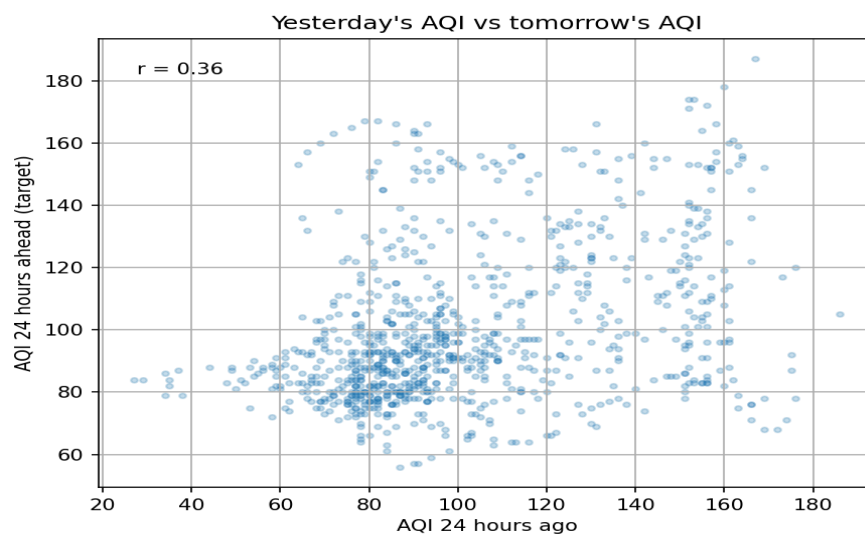
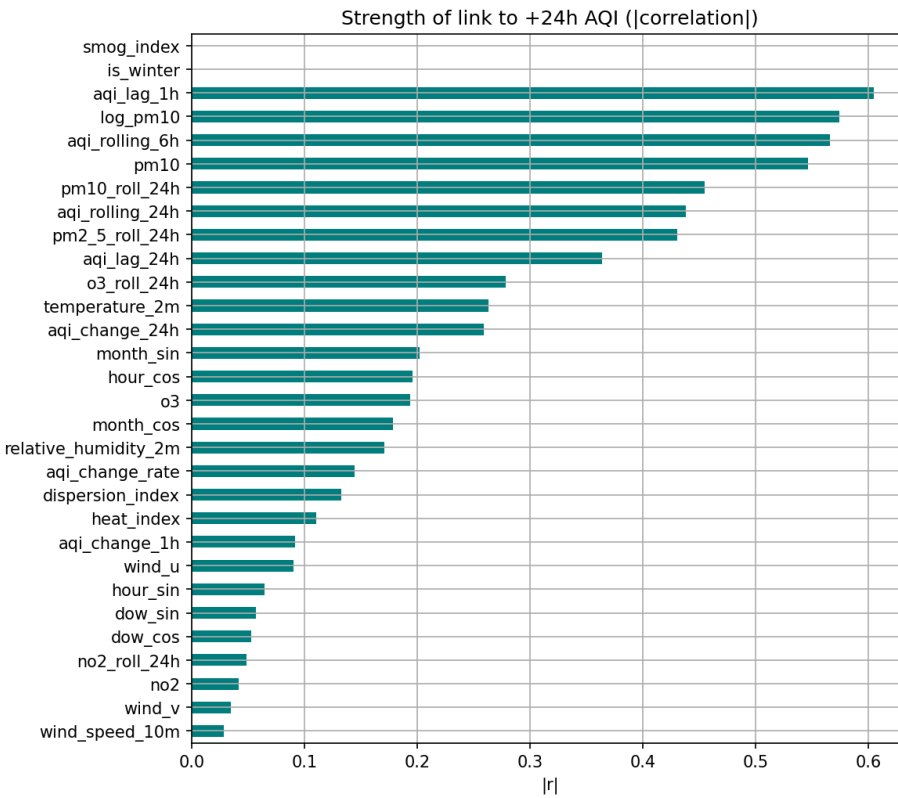


(ii): AQI peaks in early morning (06:00–09:00) and evening (18:00–22:00).



(iii): Higher wind speed correlates with lower AQI (dispersion).

Some other EDA visuals that were created:



4. Model Training and Evaluation:

The best model was found to be Random Forest. The following are the comparison metrics between the different models used:

Model	Avg RMSE	Avg MAE	Avg R²
Linear Regression	39.127	32.102	-0.104
Ridge	39.268	32.160	0.304
Random Forest	35.37	29.007	0.470
XGBoost	37.11	29.607	0.012

5. CI/CD Automation and Pipeline Used:

Two GitHub Actions workflows keep the pipeline hands-free. The Feature Pipeline runs every hour to fetch recent data and upsert MongoDB. The Training Pipeline runs daily at 02:00 UTC to retrain and register the best model. Both use MONGODB_URI from GitHub Secrets and support manual triggers via workflow dispatch.

6. Current Limitations in the Project:

(i) Even the best model is not performing as much as one would desire. This can be improved in the future by improving the quality of fetched data, boosting feature engineering and maybe by combining data from multiple APIs and increasing cleaning and preprocessing significantly.

(ii) Forecasts rely on Open-Meteo rather than independent ground stations.

(iii) Currently using only 90 days backfill so it may miss seasonal cycles. Switching to 6 months or maybe even a year of backfill data might improve model performance.

7. Issues faced throughout the Project:

(i) Hopsworks integration initially worked, but due to a cluster-side issue, GitHub Actions jobs were hanging indefinitely and no data was being updated in the feature store.

(ii) I migrated to MongoDB Atlas, but the free-tier 512 MB storage limit was exhausted after about two weeks, causing feature and training pipelines to fail until old data was removed or the plan was upgraded.

(iii) The interface has been directly deployed on streamlit cloud (free plan) so sometimes app reboots took too much loading this was also noticeable sometimes during live inference on the interface.